HARDWARE_V2 DATA TRANSFER PROTOCOL

Firmware Version	Design	Release Date	Reason for Change
1.00	Legact	December 21, 2020	first edition
1.01	Legact	December 6, 2022	Add debug mode, modify ID, clock calibration

<How the device works>

The device starts to work automatically after 5 minutes of power-on. During this period, unless interrupted by server instructions, it will continue to work and send results every second and every minute. When it is detected that the bed is in bed for 30 consecutive seconds, the recording will be started and the data of every minute will be stored in the flash. The maximum length of a single recording is 1440 minutes. When the person leaves the bed for 30 minutes, the record automatically stops.

The device generates a daily report at 9 am every day. The report generation time can be changed by parameter setting.

4byte UNIX time difference: It should be the local time minus the time at 0:00 on January 1, 1970 in standard UTC, because the device does not know the specific time zone, so it is UTC+0. Otherwise, the bottom clock of the hardware cannot match the server time, resulting in incorrect start of the scheduled task.

4byte UNIX time difference: There is no concept of time zone in the device, that is, it works in time zone 0. The set time value should be the local time of the device minus the local time difference of 0:00:00:00 on January 1, 1970. Otherwise, the underlying clock of the hardware cannot match the server time, resulting in an incorrect start of the scheduled task.

That is: if the current time is 8:00:00 on January 1, 2022, the set value is 1641024000, and the hexadecimal representation is 0x61D00A00. In the communication protocol, the low bit comes first and the high bit comes after, then it is 0x00, 0x0A, 0xD0 , 0x61 four bytes.

Notice! The UNIX timestamp directly read by most programming languages has the local time zone, which needs to be processed. For example, Beijing time in East Eighth District needs to add 28800 to the local UNIX timestamp.

You can use the https://tool.ip138.com/timestamp/ tool for comparative testing. As shown in the figure below, note that the normal time is selected as the time of the device location, and UTC must be selected as the 0 time zone.

<Server data storage process>

The data transmitted by the device every minute is entered into the database, and each device can be stored for up to 10 days;

The device generates report data by scheduled tasks every day, and the time period is 24 hours from yesterday (default 9:00 am yesterday to 9:00 am today, which can be set). It is necessary to read the device sleep conclusion data ([0x13] command query information, [0x14]] to read the display data), and read and store it in the database. If the device is in the recording state, it can be acquired after a delay.

<Communication protocol format>

TCPSocket connects to the server, long connection, and the hardware is the client.

<communication format>

The following frame definitions are followed between the hardware and the server:

Overall data structure: **[frame header 0x7D is 1byte]+[frame type symbol 1byte]+[frame length 2byte]+[ID total 10byte ASCII code]+[content: data/response]+[end 0x0D]**

The data sent by the hardware is in the mixed format of ASCII code and hexadecimal. For details, please refer to the detailed description of each instruction.

The hardware does not have the function of data packetization, that is, the server cannot make judgments based on the number of data packets.

The frame type character: from 0x01 to 0xFF. Represents a command or content, see below for details.

The frame length of 2byte is the length of the entire frame, including the frame header and the end of the frame. Unsigned 2 bytes, low order in front and high order in back.

The ID is similar: CNU2000001 format, which identifies the device. Since the device ID cannot be known when connecting to the server for the first time, when the server sends the [0x07] command, the device will not check the ID match, that is, the ID can be any 10-digit character.

一、Hardware actively sends data/command response

The data actively sent by the hardware has the following format:

1.1 Device emits frame type [0x80]: Prompt the server important information, the content of which is an ASCII string.

1.1.1 Content: [0x00]: The length or structure of the fetch instruction is incorrect

Answer: No server answer required

1.1.2 Content: [0x01]: The device is powered on and restarted.

Answer: No server answer is required. The notification will continue to be sent every 1 second until there is a response from the server.

1.1.3 Content: [0x02]+[4byte recording start date]+[22byte data length]: Add a sleep report.

Reply: If the server does not use frame type [0x14] to read data, the device will retransmit every 1 minute. A total of 3 times will be issued. The server should read the detailed content of the report as soon as possible after accepting it. If the network status is poor, it should be delayed or read regularly.

For the data format, see the instruction description of frame type [0x13].

1.1.4 Contents: [0xF0]: The device is waiting for firmware update.

Answer: None. When the device leaves the factory or fails to update the firmware, it is in this state. The notification will continue to be sent every 1 second until there is a response from the server。

1.2 The device sends out frame type [0x87]: The device sends out after 2 minutes of idle time, once per second. Alerts the server to perform tasks such as time calibration. For details, see the frame type [0x07] command.

1.3 Frame type sent by the device [0x85]/[0x86]: Sent when the device is working. For details, see the frame type [0x04] command.

<Communication command and response>

2.1 The server sends frame type [0x01]: retransmission command

Content: none

Response: **The device sends frame type [0x81]**, content: data of the past 30 seconds.

Description: Request to retransmit the past 120 seconds of data. The returned content package is 360 bytes, a total of 30 groups, each group is 12 bytes.

The structure of each group is: [serial number 1byte]+[time 4byte] +[state 1byte]+[heart rate 1byte]+[respiration rate 1byte]+[SDATA 2byte]+[PDATA 2byte]

If [time 4Byte]=0x00000000, this piece of data is invalid.

Application: Use this command when a certain data is not transmitted due to network delay or packet loss.

2.2 The server sends frame type [0x02]: retransmission command

Content: none

Response: **The device sends frame type [0x82]**, content: data for the past 30 minutes.

Description: Request to retransmit the data of the past 30 minutes. The returned content package is 450 bytes, a total of 30 groups, each group is 15 bytes.

The structure of each group is: [serial number 1byte]+[time 4byte]+[state 1byte]+[heart rate 1byte]+[respiration rate 1byte] +[body movement 1byte]+[snoring number 1byte]+[respiratory disorder number] 1byte]+[PTHD 2byte]+[TEMP 2byte]

[Time 4byte]: The time is an unsigned integer, the low order is in the front and the high order is in the back, which is equal to the difference in seconds between 0:0:00:00 on January 1, 1970 UTC time to the current local time. Note: The device timer will have a certain time error, which may not be exactly the same as the server time.

If [time 4Byte]=0x00000000, this piece of data is invalid.

Application: Use this command when a certain data is not transmitted due to network delay or packet loss.

2.3 The server sends frame type [0x03]: stop command

Content: none

Response: **Device sends frame type [0x83]**, content: None

Description: Forcibly stop working in either mode, leaving the device in standby mode. The indicator light will be blinking light. The device automatically enters the 0x20 working mode after the start timer is greater than or equal to 5 minutes. The device sends a command (excluding [0x08]: Heartbeat Reply) to reset the internal start timer.

Normally this command is not required.

2.4 The server sends frame type [0x04]: working mode setting

Command: [1byte data in hexadecimal]

Description: In standby mode, puts the device into mode. The indicator light will stay on.

[1byte data in hexadecimal] = 0x20 to enter server monitoring mode.（notice =0x30 to enter the data debugging mode.）

A.）

Return confirmation data:

(1) The device sends out frame type [0x84] + content [0x20] confirmation, and starts to work

(2) The device sends out frame type [0x84] + content [0x21] has already started before

(3) The device sends frame type [0x84] + content [0x22] The instrument fails and cannot start

Normal work emits every second:

2.4.1 **Device send frame type [0x85]:** sent in server monitor mode. A total of 12 bytes, sent every 1 second.

The content is [serial number 1byte]+[time 4byte] +[status 1byte]+[heart rate 1byte]+[respiration rate 1byte]+[SDATA 2byte]+[PDATA 2byte]

[Serial number 1byte]: The serial number is an unsigned 1-byte integer. Cycle from 0 to 119. If the server does not receive a certain data, it can send [0x01] to resend the string of data. Note: Since the effective storage is 120, each with an interval of 1 second, only data readback within 2 minutes is supported.

[Time 4byte]: The time is an unsigned integer, with the low-order bits in the front and the high-order bits behind, equal to the difference in seconds between 0:0:00:00 on January 1, 1970 UTC time to the current local time. Note: The device timer will have a certain time error, which may not be exactly the same as the server time.

[Status 1byte]: Status per second, Data=1byte unsigned integer per second. =0x01 get out of bed in this second; =0x02 move in this second; =0x03 sit up in this second; =0x04 sleep in this second; =0x05 wake up in this second; =0x06 heavy object in this second; =0x07 snore in this second; =0x08 weak breathing;

[Heart rate 1byte]: The heart rate is an unsigned 1-byte integer. Its true value = byte value. The value is 0 when out of bed or when it cannot be calculated.

[Respiration rate 1byte]: The breathing rate is an unsigned 1-byte integer. Its real value = byte value/10. The value is 0 when out of bed or when it cannot be calculated.

[2byte of SDATA]: SDATA is an unsigned 2-byte integer, with the low-order bits in the front and the high-order bits behind.

[2byte of PDATA]: PDATA1 is an unsigned 2-byte integer, with low-order bits in front and high-order bits behind.

2.4.2 Device send frame type [0x86]: sent in server monitor mode. A total of 15 bytes, issued every 1 minute.

Its structure is: [serial number 1byte]+[time 4byte]+[state 1byte]+[heart rate 1byte]+[respiration rate 1byte] +[body movement 1byte]+[snoring number 1byte]+[respiratory disorder 1byte] +[PTHD 2byte]+[TEMP 2byte]

[Serial number 1byte]: The serial number is an unsigned 1-byte integer. Cycle from 0 to 29. If the server does not receive a certain data, it can send [0x02] to resend the string of data. Note: Since the effective storage is 30, and each interval is 1 minute, only data readback within 30 minutes is supported.

[Time 4byte]: The time is an unsigned integer, with the low-order bits in the front and the high-order bits behind, equal to the difference in seconds between 0:0:00:00 on January 1, 1970 UTC time to the current local time. Note: The device timer will have a certain time error, which may not be exactly the same as the server time.

[Status 1byte]: Status per minute, Data=1byte unsigned integer per minute. =0x01 for the separation bed; ~~=0x02 for the split movement;~~ =0x03 for the sitting up; =0x04 for the sleep; =0x05 for the awake; =0x06 for the heavy object;

[Heart rate 1byte]: The heart rate is an unsigned 1-byte integer. is the final comprehensive judgment value of the minute. Its true value = byte value. The value is 0 when there is an exit in this minute.

[Respiration rate 1byte]: The breathing rate is an unsigned 1-byte integer. is the final comprehensive judgment value of the minute. Its real value = byte value/10. The value is 0 when there is an exit in this minute.

[body motion 1byte]: body motion is an unsigned 1-byte integer. Represents the number of body movements per minute.

[Number of snores 1byte]: snore is an unsigned 1-byte integer. Represents the number of snores per minute.

[Number of breathing disorders 1byte]: The breathing disorder is an unsigned 1-byte integer. Represents the number of respiratory disturbances per minute.

[PTHD 2byte]: Unsigned 2 bytes, low order in front and high order in back.

[TEMP 2byte]: Unsigned 2 bytes, low-order first and high-order last。

B. [1byte data in hexadecimal system] =0x30 to enter the data debugging mode.

Except that the device is in BLE debugging mode, otherwise it will enter immediately after receiving the modification command.

Return confirmation data: (1) The device sends frame type [0x84] + content [0x30] confirmation and starts working

(2) The device sends out frame type [0x84] + content [0x31] failure cannot start

2.4.3 Type of frame sent by the device [0xF0]: sent in debug mode. The content data totals 387bytes and is sent out every 1 second. The first 12 bytes are the same as [0x85]. The last 375 bytes, each 3 bytes represent a data point, a total of 125.

2.4.4 Type of frame sent by the device [0x86]: sent in debug mode. The content data is 15 bytes in total and is sent out every 1 minute. Equivalent to 2.4.2

For the debugging mode, the server needs to be turned on specially, unless the sending frame type [0x03] stops, it will continue to work until the end of the 24-hour recording. The received data needs to be stored in the database for future reference.

2.5 Server sends frame type [0x07]: Query device status

Command: none

Response: **The device sends frame type [0x87]**, content 6byte: [mode 1byte] + [error 1byte] + [4byte time]

Description: Query the current state of the device. Returns the mode and device state that came out.

Since the first connection to the server does not know the device ID, the ID in the sending frame is any 10-bit ASCII code and the device will respond

[Mode 1byte]=0x00 stands for standby mode;=0x20 stands for server monitoring mode;=0x30 stands for data debugging mode;=0x40 stands for BLE debugging mode;

=0xF0 means waiting for firmware update;

[Error 1byte]=0x00 means no error;=any other value means the device is faulty and needs to be returned to the factory.

Description of working mode:

In the standby state of 0x00, you can use frame type [0x0B]: calibrate the clock, then frame type [0x04]: work mode setting starts to work

In the 0x20 state, you can wait for data;

At 0x30, you can use the frame type [0x03] stop command to make it stand by, and then start working again;

In 0x40, it is recommended to wait for it to return to the 0x00 state, because in the 0x40 state, the BLE communication is at a high priority. At this time, the server sends the commands, except for the [0x07] query status, [0x08] heartbeat command and [0x0C] version query command. , other commands do not respond.

=0xF0 means waiting for firmware update; at this time, the device is in the factory state, the firmware is not loaded, and the server needs to update the firmware.

[Time 4byte]: The time is an unsigned integer, with the low-order bits in the front and the high-order bits behind. It is equal to the difference in seconds between January 1, 1970 0:00:00 and the present. Note: There will be a certain time error when receiving the device timer, which may not be exactly the same as the server time. Therefore it is best to calibrate every day.

2.6 Server sends frame type[0x08]: Heartbeat response

Command: none

Response: The device sends frame type [0x88] with no content

Note: Since the device cannot determine its own networking status, the server needs to send a heartbeat every 1 minute. If it does not receive it for more than 5 minutes, the device will restart the wifi.

2.7 The server sends frame type[0x09]: pressure calibration command

Command: none

Answer: The device sends frame type [0x89], content [0x00] calibration complete, content [0x01] can not be calibrated

Note: When using for the first time or when the device is unmanned, the server can send calibration instructions. The device will forcibly stop the current working state and start calibration. The device takes about 4 seconds to return.

2.8 The server sends frame type [0x0A]: query or parameter setting command

2.8.1 Command: [0x00] means reset parameter

Response: **The device sends frame type [0x8A]**, content [0x00] indicates that the parameter has been reset

2.8.2 Command: [0x01] means read parameter

Response: **The device sends frame type [0x8A**], content [0x01] + [64byte parameter]

2.8.3 Command: [0x02]+[64byte parameter] means setting parameter

Response: **The device sends frame type [0x8A]**, the content [0x02] indicates that the parameter has been modified, and the content [0x03] indicates that the number of data bits is incorrect or the parameter is wrong.

Description: Among them, there are 64byte parameters, each 4 bytes is a parameter, in order from Para1 to Para16, the format is float, the low order is in the front and the high order is in the back.

16 parameters:

Parameter 1: The dynamic threshold of leaving the bed, the initial value is 40, if it is higher than this value, it is considered to be in bed. The device automatically adjusts to the environment and the user. Automatically adjusted by AI algorithm, manual modification is invalid.

Parameter 2: Body movement and static threshold, the initial value is 10000, above this value is considered body movement. The device automatically adjusts to the environment and the user. Automatically adjusted by AI algorithm, manual modification is invalid.

Parameter 3: Pressure static threshold, the initial value is 150, if it is higher than this value, it is considered to be in bed. The device automatically adjusts to the environment and the user. Automatically adjusted by AI algorithm, manual modification is invalid.

Parameter 4: snoring threshold, the default value is 18.5. The smaller the value, the easier it is to judge snoring

Parameter 5: Weak breathing threshold, the default value is 5.0. The smaller the value, the easier it is to judge weak breathing

Parameter 6: Internal algorithm parameter, the default value is 2.3. do not change

Parameter 7: Internal algorithm parameter, the default value is 17.0. do not change

Parameter 8: Internal algorithm parameter, the default value is 4.0. do not change

Parameter 9: Internal algorithm parameter, the default value is 13.0. do not change

Parameter 10: Internal algorithm parameters, the default value is 3.0. do not change

Parameter 11: Internal algorithm parameter, the default value is 6.0. do not change

Parameter 12: Internal algorithm parameters, the default value is 9.0. Indicates that the report is generated at 9:00 every day. 0~23 can be set.

Parameter 13: Internal algorithm parameter, the default value is 1.0. The device is automatically adjusted, do not change

Parameter 14: Internal algorithm parameter, the default value is 1.0. do not change

Parameter 15: Internal algorithm parameter, the default value is 1.0. do not change

Parameter 16: Internal algorithm parameter, the default value is 1.0. do not change

2.9 The server sends frame type [0x0B]: calibration clock command

Command: [4byte time] means to set the clock

Response: **The device sends frame type [0x8B]**, the content is no, indicating that the device has finished setting the clock

[Time 4byte]: The time is an unsigned integer, with the low-order bits in the front and the high-order bits in the back equal to the difference in seconds between 0:0:00:00 on January 1, 1970 UTC time to the current local time. Note: The device timer will have a certain time error, which may not be exactly the same as the server time.

The device will forcibly stop the current working state and update the clock.

2.10 The server sends frame type [0x0C]: device version and firmware version query

Command: none

Response: **The device sends frame type [0x8C]**, content [multi-bit characters]

Description: Used to determine the device version and firmware version of the system.

[multi-character]: something like: "Hardware_V2_HVER100_FVER100" contains name hardware_V2, device version number 100, firmware version number 100.

Similar: "IAP for hardware.VER:100." contains the name hardware, IAP version number 100. At this time, the device is in a state of waiting for firmware upgrade. The server can initiate the firmware upgrade process.

Application: Used when the server determines the identity and whether a firmware update is required.

2.11 Server sends frame type [0x0D]: firmware update command

The firmware file name obtained by the server is hardware_V6_HVER100_FVER101_20200816.bin, and the name of the device, HVER and FVER are checked when updating is required. The bin file needs to be sub-packaged, each package is 512 bytes long, and the last package is insufficiently filled with 0xFF.

After the device receives the required update, it will forcibly stop the current working state.

2.11.1 Command: [0x00] +[2byte package number] means start updating, 2byte package number=bin file byte number divided by 512, low order in front and high order in back

Reply: **The device sends out frame type [0x8D]**, content [0x00] means it is ready. The device needs about 3 to 5 seconds to clear the storage space, and the server needs to wait.

2.11.2 Command: [0x01] +[2byte package serial number]+[512byte content package] means to update a certain package

Response: **The device sends out frame type [0x8D]**, content [0x01]+[0x00] +[2byte packet number] indicates correct reception,

Content [0x01] + [0x01] + [2byte packet number] indicates a receiving error

Content [0x01]+[0x02]+[2byte packet number] indicates device write error

The server is required to resend the packet when receiving errors. Device write errors require restarting the update process to allow the device to clear storage again.

2.11.3 Command: [0x02] Indicates that the update is complete and requires restarting the device

Response: **The device sends frame type [0x8D]**, content [0x02] + [0x00] indicates that the verification is over and ready to restart,

The content [0x02]+[0x01]+[2byte package sequence number] indicates that a package cannot be restarted if it is not completed.

After the device restarts, a restart command will be issued. The server can judge whether the update is correct according to the frame type [0x0C]: device version and firmware version query.

During the whole process, if the server does not get a response from the device, it means that the network is faulty. You can wait for 1 hour before retrying the process.

2.17 The server sends frame type [0x13]: device storage report query

Command: none

Response: **The device sends frame type [0x93]**, content [15 groups of data], a total of 390 bytes

Description: Returns the start time of the 15 groups of reports in the device's internal reports. Each group contains [4byte start time]+[22byte conclusion data].

[Time 4byte]: The time is an unsigned integer, with the low-order bits in the front and the high-order bits behind, equal to the difference in seconds between 0:0:00:00 on January 1, 1970 UTC time to the current local time. Note: The device timer will have a certain time error, which may not be exactly the same as the server time.

If all 4bytes are 0xFF, the record is invalid.

[22byte conclusion data] Each 2byte group, the low order is in the front and the high order is in the back. include:

[2byte total sleep duration]: the length of the longest sleep period in a day, in minutes. Equal to 0 or 0xFFFF, the record is invalid. All conclusions are set to zero.

[2byte sleep efficiency]: expressed as a percentage, the output is 0~100. It mainly focuses on the proportion of effective sleep time to total sleep time.

[2byte sleep quality]: The score indicates the health of sleep, and the output is 0~100.

[2byte number of turns over]: Indicates the number of turns over in the sleep data period.

[2byte sleep time]: Indicates the time from bedtime to sleep, in minutes.

[2byte number of getting out of bed]: Indicates the total number of getting out of bed from the start of bed entry to the end of sleep.

[2byte sleep rhythm phase]: Sleep health indicator.

[2byte slop1]: Reserved indicator, not used yet.

[Start time of 2bytes]: The starting minute of the longest sleep, the value is one of 0~1440, which means that the longest sleep starts at a certain minute of the total report length of 1440 minutes in the current day. The exact time needs to use the start time of the report + this value * 60 seconds.

[2bytes of breathing disorder index]: The breathing disorder index AHI during the entire sleep period

[2byte total snoring times]: The total snoring times during the entire sleep period.

In the device, the report generation of the previous day starts at 9:00 by default. After the report is generated, a notification will be sent to the server, and the server can read it immediately. Or the server periodically queries, if you need to send the detailed data of a report, send the frame type [0x14] to read the detailed information.

2.18 The server sends frame type [0x14]: device storage data read

Command: [4byte start time]+[1byte serial number 0~11]. The start time of 4byte is obtained from frame type [0x13].

Serial numbers 0~11 represent: a total of 24 hours of data, read in 12 times.

Response:

The device sends frame type [0x94], content [4byte start time]+[1byte serial number 0~11]+[data 480byte].

The device sends out frame type [0x94], content [4byte start time]+[1byte=0xFF]. Read error. Time or serial number is wrong.

Description: Returns the data of a certain 2-hour period of a report. Each 4byte of data is a group representing 1 minute of data, a total of 120 minutes of data.

4byte is:

[1byte status]:

=0x01 get out of bed; ~~=0x02 move;~~ =0x03 sit up; =0x04 sleep; =0x05 wake up; =0x06 heavy object;

=0x11 Deep (deep sleep);=0x12 Light (light sleep);=0x13 REM;=0x14 Wake (awake);=0x15 Out out of bed;=0x16 heavy object;

The state is composed of analytic and non-analytical periods, so multiple states need to be displayed. When it is indicated, 0x04 and 0x12 are displayed in the same state (light sleep), 0x05 and 0x14 are in one state; 0x01 and 0x15 are in one state; 0x06 and 0x16 are not displayed temporarily. The displayed states include: getting out of bed, sitting up, awake, REM, light sleep, deep sleep

[1byte heart rate]: Heart rate is an unsigned 1-byte integer. is the final comprehensive judgment value of the minute. Its true value = byte value. The value is 0 when there is an exit in this minute.

[1byte breathing rate]: The breathing rate is an unsigned 1-byte integer. is the final comprehensive judgment value of the minute. Its real value = byte value/10. The value is 0 when there is an exit in this minute.

[1byte body motion]: body motion is an unsigned 1-byte integer. Represents the number of body movements per minute.

2.21 server sends frame type [0x17]: clear all data

Command: None. clear all data

Response: **The device sends a frame type [0x97]** with no content. Indicates completion.

Description: Reset device data